

Does the Risk Premium Move with the Fed–Market r^* Wedge?

Gate results and one honest problem

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The question

The object. Before each FOMC meeting there is a signed gap between what the Fed says the longer-run policy rate is and what the market is pricing: $\Delta_t = r_{Fed,t}^* - r_{Mkt,t}^*$.

The decomposition. A long yield is an expectations component plus a term premium. Hillenbrand [2] shows the announcement window carries the entire secular decline in yields and attributes it to the market learning the Fed's longer-run view — an *expectations* channel. He does not decompose.

The gap. Rungcharoenkitkul and Winkler [1] build the two-belief structure this question needs, simulate yield curves in Section 6, and state in footnote 14 that the model embeds zero term premia. So the premium component is assumed away by one paper and not measured by the other.

This talk. Whether the *premium* component of the announcement-window move responds to the pre-meeting wedge. Everything below is a gate result: I ran the falsification conditions before running the test.

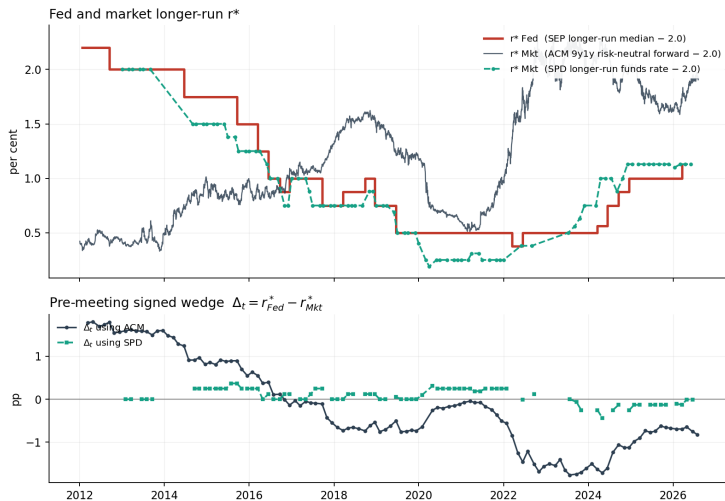
Fed side. SEP longer-run federal funds median minus longer-run PCE, 57 dot-plot meetings, January 2012 through March 2026. Participant-level where the compilations are de-anonymised, which is 36 meetings through 2020.

Market side, two candidates run side by side. The Survey of Primary Dealers longer-run funds rate, 92 surveys; and the ACM [3] 9y1y risk-neutral forward, $10 \cdot \text{RNY}_{10} - 9 \cdot \text{RNY}_9$, daily. The second is pricing-based and has the premium already stripped.

Event panel. 116 scheduled FOMC announcements, 2012–2026. The wedge is measured *strictly before* the announcement: the Fed side is the *previous* SEP, because the current one is released at the announcement itself. This is the ex-ante requirement, not a robustness check — if the announcement closes the wedge, a contemporaneous measure is endogenous by construction.

Window. Two-day change, $t - 1$ to $t + 1$, in $y^{(n)}$, its risk-neutral component and its term premium, $n = 1, \dots, 10$. The additive identity holds to 1.4×10^{-14} bp.

The two wedges



What the gate killed

Check 1, π^* dispersion: dead, and completely. All 594 participant longer-run PCE submissions across 36 meetings are exactly 2.0. Maximum within-meeting standard deviation: 0.000000. Meetings with any dispersion: zero of 36. The SPD side is identical — 2.0 in all 91 surveys, including the twelve where the value is measured rather than assumed.

→ r^* is the longer-run dot minus a constant on *both* sides. The extraction is a relabelling. The design survives; the *real-rate framing does not*, and I would rather concede that than have a referee find it.

Check 7, the SEP–SPD wedge: dead. Across 42 SEP meetings with a same-month survey, mean $|\text{gap}|$ is 13.1bp, median 13.0bp, maximum 25.0bp. Mercatus [4] reported 42bp; on this panel it is tighter still. The survey wedge has a variance of 0.028 because the two series are 94% correlated. There is no disagreement here to price.

What the gate left standing

The ACM route **survives**, with two flags.

	Check	SPD	ACM
2	$\text{var}(r_{Fed}^*)$ share; $\text{corr}(F, M)$	49.9%; +0.94	50.3%; -0.60
3	R^2 of Δ on each side alone	0.03 / 0.03	0.80 / 0.80
5	$\text{corr}(\text{dr}_{Fed}^*, \text{dr}_{Mkt}^*)$	+0.60	-0.14
6	$\text{AR}(1) \rho$ of Δ_t	+0.85	+0.98

Where the variation comes from. The ACM wedge has 38 times the variance of the survey wedge, because market pricing moves *against* the SEP. That negative correlation is the entire source of identification.

Flag one. Because F and M are negatively correlated, either one alone explains 80% of Δ — not the degenerate case check 3 was written to catch, but it makes the horse race mandatory.

Flag two, and it binds. The SEP longer-run median *changed 19 times in fourteen years*. Effective N is 19, not 116.

The result

Announcement-window response to the pre-meeting wedge, bp per pp, two-day window, 116 meetings, HAC(4).

Maturity	Total yield		Risk-neutral		Term premium	
2y	+2.25	(2.57)	+1.46	(2.53)	+0.79	(1.25)
5y	+3.34	(4.06)	+1.72	(2.71)	+1.62	(3.21)
10y	+3.14	(4.18)	+1.49	(2.84)	+1.65	(2.52)

Sign. When the Fed sits above what the market is pricing, yields rise into the window. The market moves toward the Fed. That is Hillenbrand's channel and the risk-neutral column is its predicted signature — confirmation, not news.

The new column is the third one. At 10y the premium response is slightly larger than the expectations response. The SPD wedge gives nothing anywhere, at any maturity, on any component.

Before believing it: is it a free parameter of the construction?

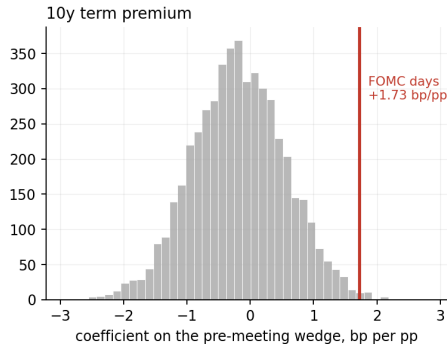
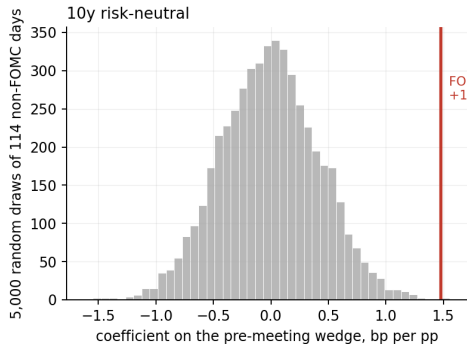
The worry. The wedge is built from the ACM risk-neutral curve at $t - 1$ and the regressand is the change in the same curve from $t - 1$ to $t + 1$. ACM fits a *stationary* VAR to yield principal components, which mechanically produces a negative level-on-change relationship. The regression could be pure mean reversion in the ACM state vector with no announcement content at all.

The test. Run the identical regression on every non-FOMC trading day in the sample — 3,513 of them.

10y component	FOMC days ($n = 114$)	non-FOMC days ($n = 3,513$)
Total yield	+3.20 (4.32)	-0.18 (-1.02)
Risk-neutral	+1.47 (2.75)	-0.01 (-0.08)
Term premium	+1.73 (2.63)	-0.17 (-1.04)

The same regression, run on 3,513 ordinary days

Placebo: the wedge does nothing on days without an announcement



Where it breaks

Inference. Moving-block bootstrap, block length eight, roughly one year of meetings. The risk-neutral coefficient gives $[+0.34, +3.01]$ and excludes zero. The term premium gives $[-0.11, +2.57]$ and **includes zero**. The effect is also purely a *level* effect — the change in the wedge since the previous meeting predicts nothing, which is check 6 biting.

Subsamples, and this is the real problem.

10y	Total		Risk-neutral		Term premium	
2012–2015, ZLB	+5.44	(2.01)	−0.48	(−0.24)	+ 5.92	(2.08)
2016–2019, normalisation	+0.22	(0.09)	−0.80	(−0.44)	+1.02	(0.76)
2020–2026, pandemic on	+6.71	(2.94)	+ 5.07	(4.00)	+1.64	(0.86)

→ The pooled headline averages two non-overlapping episodes. The premium effect is a ZLB phenomenon; the expectations effect is a post-2020 phenomenon; the middle four years are null on everything.

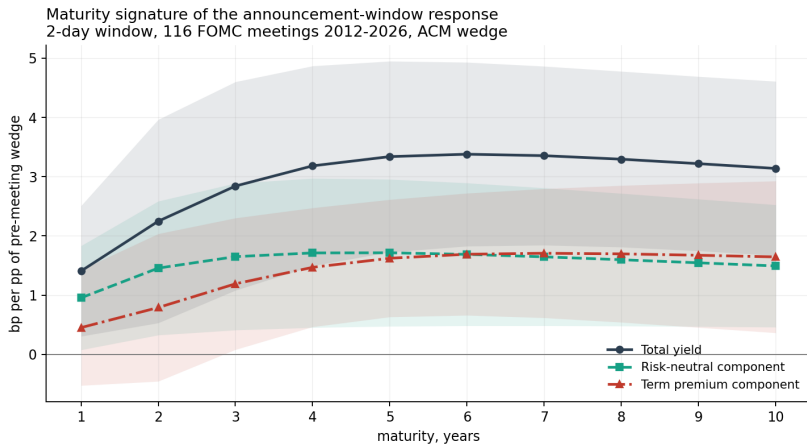
One thing that got stronger, not weaker

The premium response is not a by-product of the expectations move. Conditioning on the announcement's own near-term policy surprise — the cleanest available proxy being the window change in the 1y and 2y risk-neutral rates — the 10y term-premium coefficient *rises*.

Specification	b	t	R^2
$dTP^{(10)} \sim \Delta$	+1.65	2.52	0.039
$dTP^{(10)} \sim \Delta + drn^{(1)}$	+2.23	3.48	0.175
$dTP^{(10)} \sim \Delta + drn^{(1)} + drn^{(2)}$	+2.17	3.55	0.177

And the sign of the wedge is what prices, not its size. The absolute wedge gives nothing. Split by direction, the term premium responds *only* when the Fed sits above the market (+2.99, $t = 2.50$; the other side is a clean zero) and the risk-neutral component responds *only* when the Fed sits below (−4.04, $t = -3.96$). Two directions, two components. That is either the most interesting thing here or the same regime split of the previous slide seen twice, since 2012–2015 is a Fed-above period and 2020–2026 is a Fed-below period.

Maturity signature



Neither profile. An endpoint shock loads flat at one, a cyclical shock decays at $1/n$. Both components trace a hump peaking at 5–7y.

What I would like to ask you

One. The regime split. A premium effect that lives only at the ZLB and an expectations effect that lives only after 2020 — is that two mechanisms or one badly specified one? Everything else is downstream of this, and more robustness on the pooled sample does not address it.

Two. The Cieslak–Povala cycle. Does it still contain r^* variation, and would an external r^* measure change the risk-premium factor? Hillenbrand dismisses the risk-premium channel as unlikely to explain the *majority* of the decline, which is not the same as zero.

Three. Is anyone working on the premium component of the announcement-window move? Benigno's Substack in early August links hall-of-mirrors to Hillenbrand and to a Lustig extension, and notes that competing explanations imply different maturity profiles.

Four, a construction question. The whole premium result is one model's term premium, and ACM assumes a constant endpoint while the wedge is a statement about the endpoint. Is Kim–Wright the right adversarial test, or does its survey input make the comparison circular?

Where this leaves things

Established. The pre-meeting wedge moves announcement-window yields; the relationship is absent on 3,513 non-announcement days and sits above 99.9% of 5,000 randomised placebo samples; the risk-neutral coefficient survives block-bootstrap inference. Hillenbrand's prediction holds on this construction.

Not established. The term-premium coefficient — the actual contribution — is positive and HAC-significant, but its bootstrap interval includes zero, it is entirely a 2012–2015 phenomenon, and it exists only with the pricing-based wedge. Suggestive. Not a result.

Dead. The SEP–SPD wedge as a Fed-versus-market object. The claim to be measuring *real*-rate disagreement.

Next. Resolve the regime split; bring in a second term-premium model; move the Fed side from the median, which changed 19 times, to the cross-sectional dispersion, which takes 49 distinct values and is not near a unit root.

References

- [1] Rungcharoenkitkul, P. and F. Winkler (2026). “The Natural Rate of Interest Through a Hall of Mirrors.” *Journal of Monetary Economics* 157, 103858.
- [2] Hillenbrand, S. (2025). “The Fed and the Secular Decline in Interest Rates.” *Review of Financial Studies* 38(4), 981–1013.
- [3] Adrian, T., R. Crump and E. Moench (2013). “Pricing the Term Structure with Linear Regressions.” *Journal of Financial Economics* 110(1), 110–138.
- [4] Mercatus Center (2025). Comparison of the SEP longer-run median and the Survey of Primary Dealers, 2012–2025.
- [5] Cao, S., R. Crump, S. Eusepi and E. Moench (2020). “Fundamental Disagreement about Monetary Policy and the Term Structure of Interest Rates.” Federal Reserve Bank of New York Staff Report 934.

Appendix: the horse race

Entering the two sides separately instead of imposing the -1 coefficient the wedge assumes, 10y, two-day window.

	Wedge	Mkt alone	Fed alone	Both (Fed / Mkt)	$b_F = -b_M$
Total	+3.14	-4.59	+5.07	+3.27 / -3.00	$p = 0.935$
Risk-neutral	+1.49	-2.59	+2.07	+0.71 / -2.28	$p = 0.363$
Term premium	+1.65	-2.00	+3.00	+2.57 / -0.72	$p = 0.483$

The restriction is never rejected — the data are content to be described by a single signed wedge. But the unrestricted coefficients say something the wedge hides: the risk-neutral response loads on the *market* side and the premium response loads on the *Fed* side. The two components are answering different halves of the gap.

Appendix: effective sample size and the fallback

The median. 57 SEP meetings, 12 distinct values of the longer-run r^* , 19 meetings at which it changed, longest run without a change ten meetings. AR(1) $\rho \approx 1$.

The fallback. Cross-sectional dispersion of the same panel: mean standard deviation 0.304, range [0.217, 0.461], *49 distinct values over 57 meetings*, AR(1) $\rho = 0.803$.

→ Roughly 2.5 times the identifying variation, and not near a unit root. If the median route stalls this is the better-powered object, and it has its own asset-pricing interpretation via SR 934 [5].

Randomisation detail. 5,000 draws of 114 random non-FOMC days. Observed 10y risk-neutral coefficient sits at the 99.9th percentile ($p = 0.002$); term premium at the 99.4th ($p = 0.012$); total yield above all 5,000 draws.